

[Logo: Hackles]

Concepts Lab

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-in-inked, in the order the lessons take them.

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OCTOBER 2026

Hackles • Cybersecurity Concepts Lab for Teachers

Demo 5: The Internet as a Postal System (Packet Post)

Trust your hackles. Pause • Check • Tell

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE

Big idea	Your message is cut into numbered pieces (packets) that hop through many helpers (routers) before being reassembled. Anyone who handles a plain postcard can read it. HTTPS puts it in a sealed envelope, but the envelope still shows who it is addressed to.
Grade bands	K-2 (the pass-the-note relay), 3-5 (full Packet Post), 6-8 (full + who can see what), 9-12 (full + TLS, certificates, DNS, VPN limits)
CSTA-style concepts	Networks and the Internet: packets, routing, protocols, encryption in transit (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Index cards, envelopes, masking tape on the floor, a few string or tape "cables". No devices.
Teachers do it first	Yes. Part A about 20 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Fold a teacher-written note that says only "hello" and pass it hand to hand along one row. Who could have read it? Now seal an identical note in an envelope and pass that. Who can read it now, and what is still visible on the outside? (Never use notes students wrote themselves.)

01 The big idea

WHEN YOU SEND A message, load a picture or open a website, the data is **not** sent as one big piece along one private wire. It is chopped into small numbered **packets**, each with a "from" address, a "to" address and a number. Each packet travels through a chain of **routers** (like sorting offices), each deciding only the *next* hop. At the other end the pieces are **reassembled in order**, and any that went missing are asked for again.

This is both a powerful idea in computing (a network that keeps working when parts fail) and a **security idea**: because many strangers' machines pass your packets along, **what is written on the outside of a packet is visible to the handlers, and what is written inside is visible too unless it is encrypted**. The "lock" in your browser is the sealed envelope.

02 What a teacher should know (plain language)

2.1 THE POSTAL PARALLEL

INTERNET	POST OFFICE
Message / web page	A long letter or a big parcel
Packet	One numbered postcard of the whole thing
IP address	A street address that identifies a device's location on the network
Router	A sorting office that reads the address and picks the next road
Cables, Wi-Fi, fiber	Roads and trains
DNS (Domain Name System)	The directory that turns "school.example" into a street address
Protocol	The agreed rules (how to number, check and resend)
HTTP	Postcard: anyone handling it can read it
HTTPS (TLS)	Sealed envelope with a lock only the recipient can open
Certificate	An ID card for the website, vouched for by a trusted authority
Firewall	A guard at the sorting office door that checks what may enter

2.2 PACKETS, ROUTING AND RESILIENCE

- Packets from the same message **may take different routes** and arrive out of order. Numbers let the receiver put them back together.
- If a road is closed, routers send packets another way. That was a design goal: a network with no single point of failure.
- A receiver asks for a missing packet to be sent again (this is what the TCP protocol does; some real-time apps such as live video accept occasional loss) [VERIFY wording].

2.3 WHO CAN SEE WHAT? (THE HONEST TABLE)

HERE "VISIBLE" MEANS READABLE by someone who handles the traffic.

WHO HANDLES THE TRAFFIC	PLAIN HTTP (POSTCARD)	HTTPS (SEALED ENVELOPE)
The Wi-Fi owner or anyone who can capture that Wi-Fi's traffic	Content, pages, and any forms you type	Which website (domain) you contact, when, roughly how much data. Not the page contents or what you type.
Your internet provider	Same as above	Same as above: the site name and traffic size, not contents
Your school network and filters	Same as above, plus device logs	The site name and traffic size, and anything the school's managed device is configured to record
The website you are visiting	Everything you send to it	Everything you send to it (it is the intended recipient)
The routers in between	Content	Just the envelope: where it is going
An attacker who copies your traffic	Content	Mostly unreadable

Important nuances:

- Even with HTTPS, the **domain name** and the **amount and timing** of traffic are usually visible (the rest of the web address after the domain, and the page contents, are hidden). Technologies that hide more (such as encrypted DNS and Encrypted Client Hello) are being deployed [VERIFY current status].
- **The site itself** always sees what you send it. HTTPS is not privacy from the website (Demo 9).
- **School-managed devices and accounts** may record more than the network can see; students should assume school devices are monitored per school policy.
- **A VPN** moves who can see your traffic: the Wi-Fi owner and provider no longer see destinations; the VPN company does. It is a tradeoff, not a cloak of invisibility. Choose providers carefully [VERIFY independent audits and trustworthiness claims].

2.4 THE LOCK ICON, HONESTLY

The lock (or, in newer browsers, a different "connection settings" icon) means: **the connection between you and that site is encrypted, and the site proved it controls the domain name shown.** It does **not** mean the site is honest, safe or run by the organization you think it is. Free certificates are easy to obtain, and many phishing sites now use HTTPS (the Chromium team cited the ubiquity of HTTPS when retiring the lock icon; a specific percentage was not verified and is not claimed) [VERIFIED 2026-10-05: [blog.chromium.org](https://blog.chromium.org/2023/05/02/https-ubiquity/), 2 May 2023]. Chrome announced in May 2023 that it would replace the lock icon with a neutral "tune" icon [VERIFIED 2026-10-05: [blog.chromium.org](https://blog.chromium.org/2023/05/02/an-update-on-the-lock-icon/), "An Update on the Lock Icon"]; other browsers have also changed their symbols. Teach students to read the **domain name** carefully, not the icon: take the host name (between "https://" and the first single slash, ignoring anything before an @ sign) and read it from the right. For these examples the last two words, such as login-help.test, are the real domain; everything to their left is decoration.

2.5 WHERE DEFENSE HELPS

- Prefer HTTPS sites and apps; heed "Not secure" warnings and never ignore certificate warnings (they can mean a fake site or tampered connection).
- On open Wi-Fi, avoid typing passwords into non-HTTPS pages and keep software updated (Demo 7); use 2-step verification (Demo 6) and password managers that only fill on the right site.
- Treat names, addresses and photos as postcards: if it would be uncomfortable on a postcard, think twice.

2.6 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"Padlock means the site is safe."	It means encrypted connection and a valid certificate for that domain only.
"Incognito or private mode hides me from my school or provider."	It only stops your device from saving history locally.
"On public Wi-Fi, everything is visible."	With HTTPS, much less is. But risks remain (fake hotspots, device vulnerabilities).
"A VPN makes me anonymous."	It shifts who sees traffic and does not defend against phishing or account tracking.
"Deleting a message deletes it everywhere."	Copies may remain on servers, in backups and at recipients.

03 Materials (zero to low cost)

- Index cards (6 per team, numbered, each addressed "To: F"), envelopes and a "lock" (a lock stamp or tape), masking tape or six desks as a map of "towns", yarn "cables", a stopwatch.
- A **Snoop Clerk** badge (a silly hat, obviously fictional) and the student handout.

Domain Reading Cards: five fictional web addresses are printed on the student handout (answer-free). The ".example" and ".test" endings are reserved for documentation, so you never point at a real site. The reading of each address is on the teacher answer page.

04 Part A: Teachers do it first (about 20 minutes)

YOU NEED 6 TEACHERS or helpers (or use all 6 roles with some doing two).

1. **Set up the map (4 minutes).** Mark six "towns" (A to F). Connect A to B and C; B to D and E; C to E; D to F; E to F. Roles: Sender at A, Receiver at F, Routers at B, C, D, E. Make the connections one-way arrows or yarn.
2. **Send a message (6 minutes).** The Sender writes a short sentence on six cards, one word per card, numbered 1-6, each addressed "To: F". Each router, seeing only "To: F", picks a next hop (rule: pick the closer link; if closed, the other). One road may be "closed" by the teacher mid-game. The Receiver collects, reorders and requests any missing card.
3. **Add the Snoop Clerk (4 minutes).** A Snoop Clerk stands at router E (close the road from B to D so every packet passes E). For plain cards (postcards) they read the words aloud ("Hmm, 'Meet me at the arcade'"). Discuss what can be learned from just the outside of the card.
4. **Seal it (4 minutes).** Repeat with sealed envelopes: the sender writes the word in a shift cipher (key agreed in secret, Demo 4) or simply seals it; Snoop Clerk sees the "To" and "From" and the weight, not the word.
5. **Reflect (2 minutes).** What did you learn about "who can read what"? What would a website need to prove to you that you're sending to the real one (certificates)?

05 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook: pass a teacher-written blank note across the class and ask who could read it (everyone who touched it).
2-4	Introduce roles and the map.
4-10	Run Packet Post once with postcards (5 cards, 4 routers). One road closed at minute 7.
10-13	Run the sealed-envelope round. Snoop Clerk tries to read.
13-15	Exit slip: "Who can read a postcard? Who can read a sealed envelope? What is still visible?"

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-5	Hook and the postal parallel table.
5-17	Packet Post round 1 (postcards), count time; add a closed road; add a dropped packet and a resend request.
17-27	Round 2: sealed envelopes. Snoop Clerk reports what is still visible (who, when, how big).
27-35	"Is the lock enough?" Domain Reading Cards: students decide the real destination and discuss the HTTPS-but-fake idea.
35-42	"Who can see what?" table sort: students place cards (Wi-Fi owner, school, website, VPN) in a grid for postcard vs envelope.
42-45	Exit ticket and handout.

VARIATION FOR K-2 (5-8 MINUTES)

"Pass the Note": pass a teacher-written note hand to hand and notice everyone who touched it. Then wrap it in a folded paper "envelope" with a drawn lock: "Who can read it now?" Message: share personal things only in a sealed, trusted place, and ask a grown-up first.

06 Teacher script

Hook (2 minutes). "When I press send, how does my message get across the world in a blink? It doesn't go on one road. It turns into numbered postcards that each hop through helpers. Today you're going to be those helpers. There's a catch: helpers can read postcards."

During the round. "Router, you only know where to send the next hop. Receiver, if you're missing number 4, what do you do? Ask for it again. That's how a bad connection still gets the whole picture."

Snoop Clerk moment. "Our Snoop Clerk reads every postcard. Are they a villain? Maybe not. Any device that handles a postcard can read it. It's nothing personal. That's why we use sealed envelopes, called HTTPS."

On the lock. "The padlock means the envelope is sealed and addressed to the real domain. It doesn't mean the person at the other end is honest. A scammer can also buy a sealed envelope. Read the address."

Close. "Packets, routers, protocols, envelopes. You now know what happens to a message you send. When something about it feels off, a strange website, a warning in your browser, pause. Check the address, ask a grown-up, tell."

07 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	Pass the note across the classroom; a sealed folded note with a lock drawing. "Everyone who touches a note can read it unless it's sealed."
3-5	Packet Post with numbered postcards; closed road; resend; sealed envelope round.
6-8	The postal system with a Snoop Clerk. Who can see what: Wi-Fi, ISP, school, site. Read the domain. Discuss VPNs as "mail forwarding" with its own tradeoffs.
9-12	TLS handshake, certificates, DNS, VPN limits. Public key crypto to agree on a session key (Demo 4), certificate authorities as notaries, domain validation vs identity, metadata visibility.

08 Discussion questions

1. Why is it helpful that packets take different routes? What risk does it create?
2. If you were the Snoop Clerk at a router, what could you learn from the outside of the envelope alone?
3. What does the padlock tell you, and what does it NOT tell you? Why can a fake site have HTTPS?
4. (9-12) How does the browser know which public key belongs to which website?

09 Level-up challenge

- **Trace a route (teacher demonstration only).** Show a printed or recorded example of a route trace for a general site and compare it with the class map. Students do not run tracing or scanning tools themselves [VERIFY that showing one is allowed under school policy; skip if not].
- **Design a protocol.** Write rules for what the Receiver does if packet 3 never arrives, arrives twice or arrives corrupted (add a "checksum": the number of letters on the back).
- **Layered envelopes (11-12).** Add "onion routing" ideas: three envelopes within envelopes [VERIFY].

10 Safety and ethics notes (defense only)

- The Snoop Clerk is a **role**, not a how-to. Do not teach packet sniffing, spoofing, or interception of any real traffic. Students must never capture or read other people's network traffic. That is illegal without authorization (see Demo 14).
- Do not run scanning or tracing tools against school networks or anyone's devices.
- Do not ask students to disclose their home network details, Wi-Fi names or passwords.
- Keep the Snoop Clerk silly and friendly. Do not use a scary "hacker in a hoodie" figure.
- Tell students that school devices and networks may be monitored per school policy, in a neutral, factual tone.

11 Assessment idea

- **Exit ticket.** "Write one thing a sealed envelope hides and one thing it does not hide." Rubric: 3 = content hidden and domain/size/timing or recipient visible; 2 = content hidden only; 1 = "everything is hidden".
- **Domain detective.** Give five fictional URLs; students circle the real domain and say whether to proceed. Include one with a padlock but a fake domain.

12 Sources and [VERIFY] list

- CSTA 2017 Standards, 1B-NI-04 (packets) and 2-NI-04 (protocols) [VERIFIED 2026-10-05: CSTA K-12 CS Standards (2017, rev.) text read; mapping approximate].
- IETF RFCs for TCP (RFC 9293) and TLS 1.3 (RFC 8446) [VERIFIED 2026-10-05: rfc-editor.org].
- Chrome's replacement of the lock icon with a "tune" icon (announced 2 May 2023) and Google's reasoning [VERIFIED 2026-10-05: blog.chromium.org].
- Anti-Phishing Working Group reports of HTTPS use on phishing sites [percentage not verified].
- Encrypted DNS (DoH, RFC 8484; DoT, RFC 7858 [VERIFIED 2026-10-05: rfc-editor.org]) and Encrypted Client Hello status [ECH deployment status not checked].

- CS Unplugged (CC BY-SA 4.0, VERIFIED 2026-10-05) and Code.org "Internet" lessons (packets and routing) as inspiration [Code.org lesson specifics not checked].

SECTION

Student Handout: Packet Post

Name: _____

Words to know

- **Packet:** a small numbered piece of a message.
- **Router:** a helper that sends packets to the next stop.
- **Protocol:** the rules everyone agrees to follow.
- **HTTPS:** a sealed envelope for web traffic.

Draw it

DRAW THE MESSAGE "HELLO FRIEND" cut into packets. Number them. Draw two different paths they could take to your friend.

Postcard or sealed envelope?

SITUATION	WHO CAN READ THE MESSAGE?
Postcard (HTTP)	_____
Sealed envelope (HTTPS)	_____
What is still visible on the outside?	_____

Read the address!

HOW TO READ ONE: take the host name (between "https://" and the first single slash, ignoring anything before an @). Read it **from the right**. The last two words are the real domain. Circle it, then write SAFE-LOOKING or PAUSE.

1. https://www.myschool.example/library _____
2. https://myschool.example.login-help.test/signin _____
3. http://school-portal.test/login _____

4. <https://www.myscho0l.example> _____
5. <https://myschool.example@evil.test/page> _____

Think about it

THE PADLOCK MEANS THE connection is _____. It does **not** mean the site is _____.

Remember

- Packets can take different routes and still arrive.
- A padlock is not a promise of honesty. Read the address.
- Never type a password on a page without a lock, and don't ignore browser warnings.
- Your school may be able to see what you do on school devices and networks.
- If something online feels off, **Pause. Check. Tell.**

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 5 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Postcard or envelope table: postcard (HTTP) = everyone who handles it; sealed envelope (HTTPS) = only the intended receiver (the website); still visible on the outside = who it is for (the domain), when, and roughly how big.

Domain Reading Cards

URL	READING
https://www.myschool.example/library	Domain is myschool.example. HTTPS, but "fine" depends on whether this is the real domain (check it against what you know). Reasonable to proceed if it matches.
https://myschool.example.login-help.test/signin	Domain is login-help.test . The first parts are decoration. PAUSE.
http://school-portal.test/login	No lock: HTTP only. PAUSE; do not type a password.
https://www.myscho0l.example	Look-alike: a zero instead of the letter o. PAUSE and check.
https://myschool.example@evil.test/page	Everything before the @ is decoration; the real destination is evil.test. PAUSE. [VERIFY that modern browsers warn on this]

Think about it: the padlock means the connection is **encrypted** (to a site that proved it controls that domain). It does not mean the site is **honest or safe**.

Notes: the "read from the right" rule works for these examples; real domain names with endings such as .co.uk have an extra part, so teach it as a first check, not a complete test. Preferring the bookmark or the app is the stronger habit.